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Mitigating False Negatives in Neural Network Object Detectors through Light Informed Shape Analysis (LISA)

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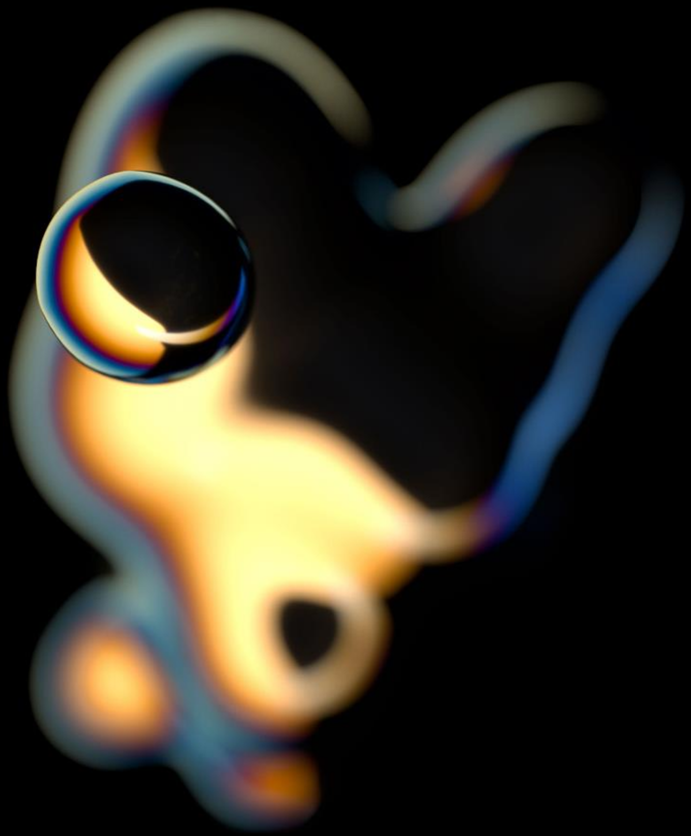
Outline



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Problem Statement

- Neural-network based (NNB) computer vision algorithms are good at solving the average case of detecting well-defined and visible objects.
- When confronted with outliers – due to inevitable changes in the configuration and/or partial concealment of the detected object - they fail and generate what's called a *false negative.*
- That is **the NNB algorithm asserts that something is not there when in fact it is present**.
- This fact is partially due to the size and scope of the training set the NNB algorithm was built upon
- **Our work is meant to provide a mapping which corrects this lack of specialized training and thus overcome and mitigate false negatives.**

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Exceeding the Training Set Boundaries Through Elastic Shape Analysis (ESA)

- The marker indicating whether AGI or ASI (Artificial General and Super Intelligence, respectively) exists concerns the ability of an AI to generate inferences that go beyond their explicit training.
    - Having an AI's reasoning confined to its training set is known as the “frame problem”.
- If an AGI or ASI requires exotic hardware to perform their inference duties, their practical deployment is limited since it cannot be migrated to small devices.
- Hence, we are advancing the notion that through ESA, we can exceed the boundaries of a computer vision’s training set and thus enhance a CV system’s “deploy-ability” due to ESA’s low size, weight and power.
- ESA requires only the use of geometric invariants to perform its classification work and thus need not have a preset or fixed training set; it need only have a vague shape prior as we will see.
- Algorithms responsible for implementing ESA can be found in [1].

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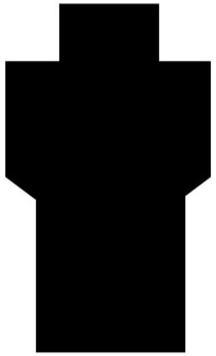
Elastic Shape Analysis (ESA) in a Nutshell

1

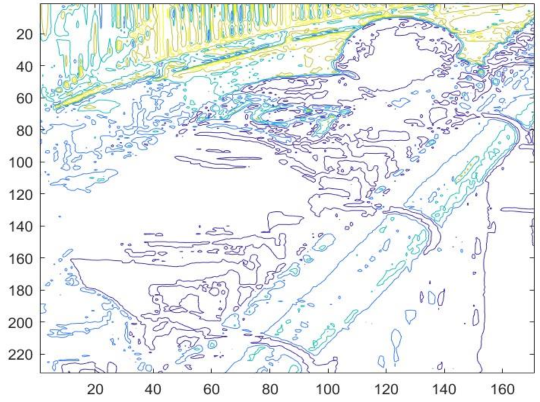


Training Set

2

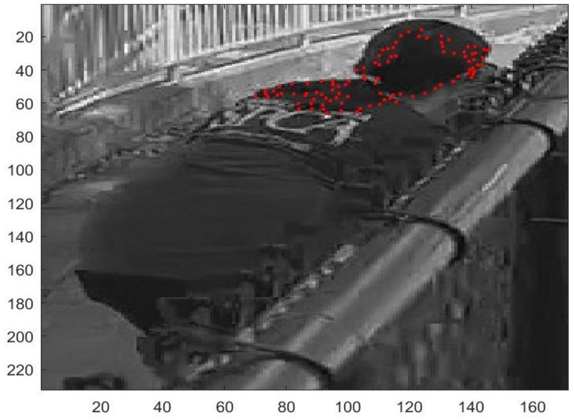
ROI Imagery

3



Contour Extraction

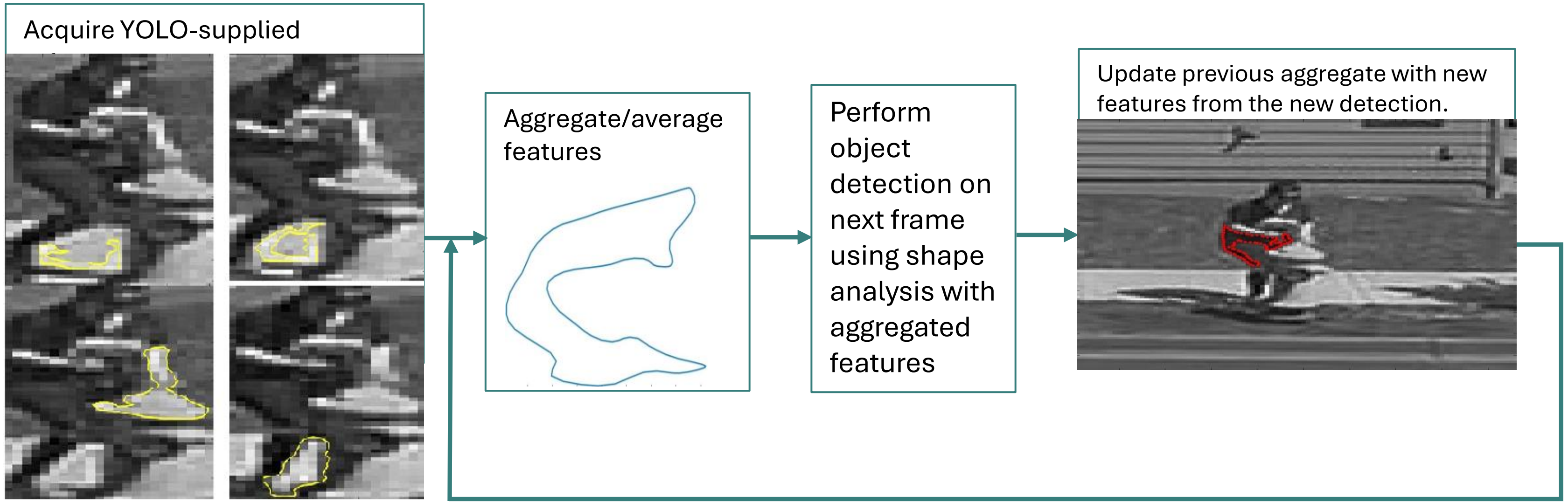
4



Target Detection

Light Informed Shape Analysis (LISA)

- LISA's inspiration comes from the idea that the best dataset for object detection and classification is reality *as it is presented*.
  - Neural nets are good at detecting/classifying objects, but they are limited by their predetermined training dataset.
- LISA's **patent pending** technique uses YOLO to supply reference contours (or shape priors), and subsequent video frames supply candidate contours.
- As new information arrives, new updates are made to LISA's understanding of the detected object's position/pose by means of understanding how light plays upon it.



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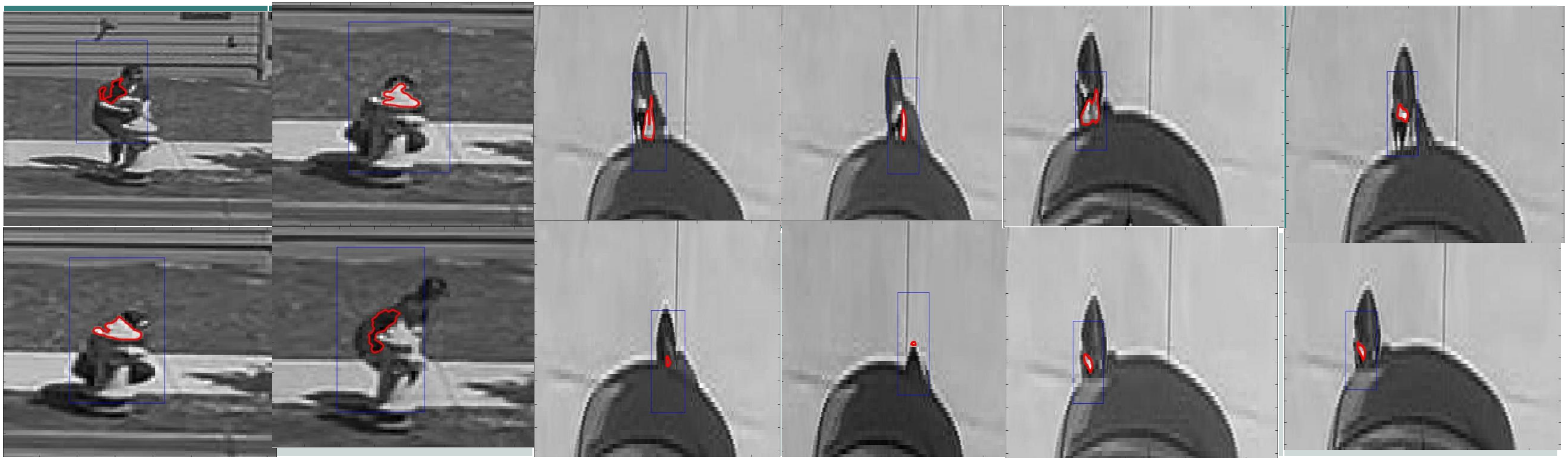
Results

- These videos are the result of LISA applied to Lusine engaged in deliberate concealment/configuration changes.
- We “pick things up” where YOLOv5 left off and thus provide a higher MAP (Mean Average Precision)



Feeling (Af)fine

- When encountering adverse imaging circumstances produced by the imaging environment (due to chain link fence, grates, grills, fog, etc.) affine transforms can be applied.

- Affine transforms allow us to accommodate “stretching” in the detected object.

Key Takeaways and Closing Thoughts

- The average case for neural network-based computer vision is largely solved.
- The statistical "long tail", however, remains open to exploration.
- LISA serves as a kind of "referee" that can offer further distinctions in the classification of a particular object.
- Adverse imaging circumstances and/or object placement can be overcome by means of a clever application of advanced mathematics.
- Discovering the initial parameters necessary to produce good outcomes is a subject of research.

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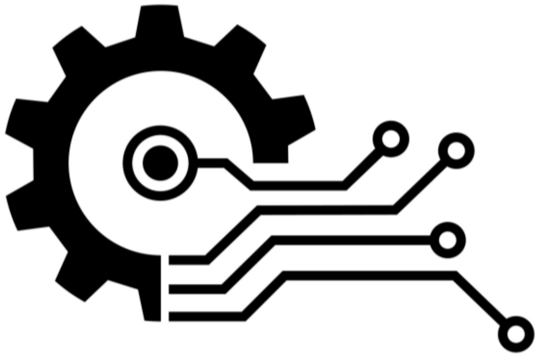
Patents Filed So Far



1. Systems and Methods for Mitigating False Negatives in Neural Network-based Object Detection Using Light Informed Shape Analysis (LISA) **(Filed)**

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References

[1] A. Srivastava and E. Klassen, Functional and Shape Data Analysis, Springer-Verlag, New York, 2016.

